

Overview

Artificial Intelligence (AI) is a branch of computer science that enables computers and machines to perform tasks that normally require **human intelligence**. These tasks include learning, reasoning, problem-solving, understanding language, recognizing images, and making decisions.

Definition

Artificial Intelligence is the simulation of human intelligence in machines that are programmed to think, learn, and make decisions like humans.

Examples of AI

- Image Recognition (Face Unlock on smartphones)
- Speech Recognition (Google Assistant, Siri)
- Chatbots (Customer Support Bots)
- Recommendation Systems (Netflix, Amazon, YouTube)
- Self-driving Cars
- Fraud Detection in Banking

Real-Life Example

When you unlock your phone using **Face ID**, AI recognizes your face by comparing it with stored facial data.

2. Types of Artificial Intelligence

A. Narrow AI (Weak AI)

Designed to perform one specific task.

Examples

- Google Translate
- Siri
- Alexa
- ChatGPT
- Spam Email Filter

B. General AI (Strong AI)

AI that can perform any intellectual task like a human.

- Still under research
- Does not yet exist

C. Super AI

An advanced AI that would surpass human intelligence in all areas.

- Theoretical concept
- Not yet developed

3. What is Generative AI?

Generative AI (GenAI) is a type of Artificial Intelligence that can create **new content** instead of only analyzing existing data.

It learns patterns from large datasets and generates human-like outputs.

Definition

Generative AI is an AI technology capable of generating new text, images, audio, videos, code, and other content based on user prompts.

Types of Content Generated

- Text
- Images
- Audio
- Video
- Music
- Source Code
- Presentations

Examples

- ChatGPT
- GitHub Copilot
- Google Gemini
- Claude
- DALL·E
- Midjourney

Real-Life Example

A student asks:

"Write a Python program to calculate factorial."

Generative AI generates the complete program instantly.

4. Difference Between AI and Generative AI

Artificial Intelligence	Generative AI
Performs intelligent tasks	Creates new content
Focuses on analysis and decision-making	Focuses on content generation
Recognizes patterns	Generates text, images, code, etc.
Example: Face Recognition	Example: ChatGPT

5. Large Language Models (LLMs)

A **Large Language Model (LLM)** is an AI model trained on **billions of words** collected from books, articles, websites, and other text sources.

LLMs understand language, answer questions, summarize information, translate languages, and generate human-like responses.

Definition

A Large Language Model is a deep learning model trained on massive text datasets to understand and generate natural language.

Popular LLMs

- GPT (OpenAI)
- Llama (Meta)
- Gemini (Google)
- Claude (Anthropic)
- Mistral

Capabilities

- Answer questions
- Write code
- Summarize documents
- Translate languages
- Generate emails
- Create stories
- Explain concepts

6. Prompt Engineering

Prompt Engineering is the process of writing clear and effective instructions (prompts) to obtain better responses from AI models.

Definition

Prompt Engineering is the practice of designing well-structured prompts to improve the quality and accuracy of AI-generated responses.

Characteristics of a Good Prompt

- Clear
- Specific
- Detailed
- Context-rich
- Goal-oriented

Poor Prompt

Explain AI.

Better Prompt

Explain Artificial Intelligence in simple language with examples suitable for beginners.

The second prompt produces a more useful response because it includes context and the target audience.

7. Tokens

A **token** is the basic unit of text that an AI model processes. A token can be:

- A word
- Part of a word
- A punctuation mark
- A number
- A special symbol

AI models read and generate responses **token by token**.

Example

Sentence:

Artificial Intelligence is amazing.

Possible tokens:

- Artificial
- Intelligence
- is
- amazing
- .

The exact tokenization depends on the model.

Why Tokens Matter

- Determine the amount of text an AI can process.
- Affect response length and cost in many AI APIs.
- Influence model performance on long documents.

8. Embeddings

Embeddings convert text into numerical vectors that capture the meaning of the text.

Texts with similar meanings produce similar vectors.

Definition

Embeddings are numerical representations of text used to measure semantic similarity between words, sentences, or documents.

Example

These sentences have similar meanings:

- "I love programming."
- "Coding is my passion."

Their embeddings will be close together in vector space.

Applications

- Semantic Search
- Recommendation Systems
- Question Answering
- Document Retrieval
- Duplicate Detection
- Clustering

9. Vector Database

A **Vector Database** stores embeddings and allows fast similarity searches.

Instead of searching for exact keywords, it finds documents with similar meanings.

Popular Vector Databases

- Pinecone
- ChromaDB
- FAISS
- Milvus
- Weaviate

Example

A user searches:

"How do I reset my password?"

The vector database retrieves documents about:

- Password recovery
- Forgot password
- Account reset

even if those exact words are not used.

10. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) combines a retrieval system with a Large Language Model.

Instead of answering only from its training data, the LLM first retrieves relevant information from external documents and then generates an answer based on that information.

Definition

RAG is an AI technique that retrieves relevant information from external knowledge sources before generating a response.

RAG Workflow

1. User asks a question.

2. The question is converted into an embedding.
3. The embedding is used to search a vector database.
4. Relevant documents are retrieved.
5. The retrieved content is provided to the LLM.
6. The LLM generates a response using that context.

Benefits of RAG

- More accurate answers
- Reduces hallucinations
- Uses up-to-date information
- Supports organization-specific knowledge
- Improves reliability

Real-Life Example

A company chatbot retrieves information from its latest HR policy documents before answering employee questions.

11. AI Agents

An **AI Agent** is an intelligent system that can perform tasks autonomously to achieve a goal.

Unlike a simple chatbot, an AI agent can:

- Understand objectives
- Plan tasks
- Make decisions
- Use external tools
- Perform multi-step workflows
- Adapt based on results

Example

A travel planning AI agent can:

1. Search for flights.
2. Compare hotel prices.
3. Create an itinerary.
4. Send booking information.

12. Responsible AI

Responsible AI ensures that AI systems are developed and used ethically, safely, and fairly.

Principles

Fairness

Avoid bias and discrimination.

Transparency

Explain how AI makes decisions.

Privacy

Protect user data and personal information.

Security

Prevent misuse and unauthorized access.

Accountability

Developers and organizations should take responsibility for AI outcomes.

Importance

Responsible AI builds trust and helps ensure AI benefits society while minimizing risks.

13. Applications of Generative AI

Generative AI is used in many industries.

Customer Service

- AI Chatbots

- Virtual Assistants

Software Development

- Code Generation
- Bug Detection
- Code Completion

Education

- Personalized learning
- Question generation
- Assignment assistance

Healthcare

- Medical report summarization
- Drug discovery support
- Clinical documentation

Business

- Email drafting
- Report generation
- Meeting summaries

Content Creation

- Blog writing
- Social media posts
- Marketing content
- Image generation

Entertainment

- Story writing
- Music generation
- Video creation

- Game content

14. Advantages of Generative AI

- Saves time
- Improves productivity
- Automates repetitive tasks
- Generates high-quality content quickly
- Supports creativity
- Provides personalized responses
- Assists in software development
- Enhances customer support

15. Limitations of Generative AI

- May generate incorrect information (hallucinations)
- Can reflect biases present in training data
- Requires high-quality prompts for best results
- Raises privacy and copyright concerns
- Does not truly understand information like humans

16. Real-Time Example

AI-Based College Assistant

A student asks:

"Explain Object-Oriented Programming with C# examples."

The system:

1. Converts the question into an embedding.
2. Searches the college notes stored in a vector database.

3. Retrieves relevant notes.
4. Sends the notes to an LLM.
5. Generates a clear explanation with examples.

This is a practical example of **Retrieval-Augmented Generation (RAG)**.